



Advanced Time Series Analysis and Forecasting

Seminar 12: Machine learning and deep learning for time series



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ **AI:** Artificial Intelligence
- ▣ **AR:** AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARX:** AutoRegressive model with eXogenous variables
- ▣ **CV:** Cross-Validation
- ▣ **DM:** Diebold–Mariano (test)
- ▣ **ENTSO-E:** European Network of Transmission System Operators for Electricity
- ▣ **EU:** European Union
- ▣ **FTSE:** Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ **GW:** Giacomini–White (test of conditional predictive ability)
- ▣ **HAR:** Heterogeneous AutoRegressive (model)
- ▣ **HICP:** Harmonised Index of Consumer Prices
- ▣ **HLN:** Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- ▣ **HU:** Hungary (Eurostat country code)
- ▣ **LSTM:** Long Short-Term Memory (network)

Acronyms used in this material (2/2)

- ▣ **MAE**: Mean Absolute Error
- ▣ **MASE**: Mean Absolute Scaled Error
- ▣ **MCS**: Model Confidence Set
- ▣ **MLP**: MultiLayer Perceptron
- ▣ **MW**: MegaWatt
- ▣ **OLS**: Ordinary Least Squares
- ▣ **OOS**: Out-Of-Sample
- ▣ **OWA**: Overall Weighted Average (of sMAPE and MASE relative to Naive 2)
- ▣ **QLIKE**: Quasi-LIKElihood loss
- ▣ **QRF**: Quantile Regression Forest
- ▣ **RMSE**: Root Mean Squared Error
- ▣ **RNN**: Recurrent Neural Network
- ▣ **RV**: Realised Variance
- ▣ **SE**: Standard Error
- ▣ **SHAP**: SHapley Additive exPlanations
- ▣ **SPA**: Superior Predictive Ability (test)
- ▣ **TCN**: Temporal Convolutional Network

Today's question and route

- **Question:** how do we validate, fit and compare machine-learning and deep models on time series without fooling ourselves?
 - ▶ this seminar comes **before** Lecture 12: the section “Prerequisites for Today” gives every definition the tasks use
- **Route**
 - ▶ Part A: lasso, ridge, adaptive lasso and elastic net by hand; the variance of bagging; boosting as gradient descent
 - ▶ Part A (continued): gradients in an RNN and an LSTM; attention by hand; DLinear as one linear map
 - ▶ Part B: cross-validation on realised variance; local and global models of EU inflation; quantile forests and monotone boosting for Romanian load; LSTM against HAR; N-BEATS on M4 hourly series
 - ▶ Part C: deep learners against HAR for crypto-assets; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	lasso and ridge under an orthonormal design; adaptive lasso and elastic net	Solved, Proposed	A1
A3, A4	the variance of bagging; boosting as gradient descent	Solved, Proposed	A3
A5, A6	gradients through a scalar RNN; the LSTM forget gate	Solved, Proposed	A5
A7, A8	attention by hand; DLinear as one linear map	Solved, Proposed	A7
B1, B2	validation of HAR; a random forest with a time index	Solved, Proposed	B1
B3, B4	local and global AR for EU inflation; the panel lasso for Romania	Solved, Proposed	B3
B5, B6	QRF and ARX for the evening peak; monotone boosting	Solved, Proposed	B5
B7, B8	LSTM against HAR; DLinear and N-BEATS on 100 M4 series	Solved, Proposed	B7
C1, C2	deep learners for crypto volatility; what is wrong in an AI answer?	Proposed	B7, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
RV of FTSE 100, DAX	Heber et al. (2009)	daily, 5-minute returns	2000–2022
HICP inflation, 27 EU countries	Eurostat	monthly	2000–2026
Romanian load	Energy-Charts (ENTSO-E)	hourly	2023–2026
M4 hourly series	Makridakis, Spiliotis and Assimakopoulos (2020)	hourly	–
RV of Bitcoin, Ether	Binance one-minute prices	daily	2018–2026

- ▣ RV in %², load in GW (MW in the answers)
- ▣ Functions in the notebook
 - ▶ `cv_estimates`, `global_local`, `penalised_fit`, `QRF`, `rv_forecasts`, `m4_global`, `dm_test`, `mcs`
- ▣ No account or key is needed

Prerequisites for Today (1/8): validation schemes

- Lag matrix: rows $(y_t; y_{t-1}, \dots, y_{t-p})$, the target and its p lags
 - ▶ **random K-fold**: rows assigned to K folds at random; **blocked K-fold**: contiguous folds
 - ▶ **purge**: drop h training rows on each side of a test block; **OOS**: test only on the last block
- Bergmeir, Hyndman and Koo (2018): random K-fold is valid when the model errors are **uncorrelated** (a well-specified autoregression)
 - ▶ it fails for time-index features and for overlapping targets, whose neighbours are almost the same number
- Overlapping h -period target and its autocorrelation

$$z_t = \frac{1}{h} \sum_{j=1}^h y_{t+j}, \quad \text{corr}(z_t, z_{t+1}) \geq \frac{h-1}{h}$$

- ▶ the bound holds with equality when y is white noise; with $h = 20$ it is 0.95

Prerequisites for Today (2/8): local, global and HAR models

- **Local** model: one per series; **global** model: the same function for all series [Montero-Manso and Hyndman \(2021\)](#)
- HAR [Corsi \(2009\)](#) for $y = \log RV$: daily, weekly and monthly means of the past

$$y_{t+h} = \beta_0 + \beta_d y_t + \beta_w \bar{y}_t^{(5)} + \beta_m \bar{y}_t^{(22)} + u_{t+h}$$

- ▶ $\bar{y}_t^{(5)}, \bar{y}_t^{(22)}$: means over the last 5 and 22 days; h : horizon; estimated by OLS, one regression per horizon (direct forecasts)

Prerequisites for Today (3/8): penalised regression

- With orthonormal predictors and OLS coefficient z_j , each penalty acts coefficient by coefficient

$$\text{lasso: } S(z_j, \lambda) = \text{sign}(z_j)(|z_j| - \lambda)_+, \quad \text{ridge: } \frac{z_j}{1 + \lambda}$$

- ▶ λ : penalty; $(u)_+ = \max(u, 0)$; S : soft thresholding [Tibshirani \(1996\)](#): small coefficients become exactly zero

- Variants

- ▶ **adaptive lasso** [Zou \(2006\)](#): threshold $\lambda/|z_j|^\gamma$, $\gamma > 0$: large coefficients are shrunk less
- ▶ **elastic net** [Zou and Hastie \(2005\)](#): $S(z_j, \lambda\alpha)/(1 + \lambda(1 - \alpha))$, $\alpha \in [0, 1]$ mixes lasso ($\alpha = 1$) and ridge ($\alpha = 0$)

Prerequisites for Today (4/8): ensembles of trees

- Bagging [Breiman \(1996\)](#): the variance of an average of B correlated predictors

$$\text{Var}\left(\frac{1}{B} \sum_b \hat{f}_b\right) = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{B}$$

- ▶ σ^2 : variance of one predictor; ρ : correlation between two predictors; more trees cannot go below $\rho\sigma^2$

- Gradient boosting [Friedman \(2001\)](#): add small trees fitted to the negative gradient of the loss

$$F_m = F_{m-1} + \nu h_m, \quad h_m \approx -\frac{\partial \ell}{\partial F} \Big|_{F_{m-1}}$$

- ▶ ν : learning rate; squared loss: h_m fits the residuals; pinball loss: h_m fits $\tau - 1\{y < F\}$
- ▶ **QRF** [Meinshausen \(2006\)](#): weights $w_i(x)$ of the training targets in the leaves that contain x ; quantiles of the weighted sample

Prerequisites for Today (5/8): recurrent networks

- RNN: the gradient to a past state is a product of Jacobians

$$h_t = \tanh(W h_{t-1} + U x_t), \quad \frac{\partial h_T}{\partial h_t} = \prod_{k=t+1}^T \text{diag}(1 - h_k^2) W$$

- ▶ h_t : hidden state; x_t : input; W , U : weights; vanishing gradients when $\|W\| < 1$ [Bengio, Simard and Frasconi \(1994\)](#)

- LSTM [Hochreiter and Schmidhuber \(1997\)](#): an additive memory cell controlled by gates

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad f_t = \sigma(\cdot + b_f)$$

- ▶ f_t , i_t : forget and input gates in $(0, 1)$; $\tilde{c}_t$: candidate; $\odot$: elementwise product; b_f : forget bias; along the cell $\partial c_t / \partial c_{t-1} = \text{diag}(f_t)$

Prerequisites for Today (6/8): attention, TCN, DLinear, N-BEATS

- Attention [Vaswani et al. \(2017\)](#): a weighted average of the values V , with weights from query-key similarity

$$\text{softmax}\left(\frac{qK'}{\sqrt{d_k}}\right)V$$

- ▶ q : query of one token; K : keys of all tokens; d_k : key dimension; softmax turns scores into weights that sum to 1
- TCN: kernel k , ℓ levels of two dilated convolutions: receptive field $1 + 2(k - 1)(2^\ell - 1)$ steps
- **DLinear** [Zeng et al. \(2023\)](#): trend = Ax (A : moving-average matrix), forecast = $W_s(x - Ax) + W_tAx$
 - ▶ x : the input window; W_s , W_t : linear maps for the remainder and the trend; **N-BEATS** [Oreshkin et al. \(2020\)](#): stacked blocks, each removing its backcast from the input

Prerequisites for Today (7/8): losses

- QLIKE [Patton \(2011\)](#) for a variance forecast F , and the pinball loss for a quantile

$$\text{QLIKE} = \frac{RV}{F} - \log \frac{RV}{F} - 1, \quad \rho_{\tau}(u) = \max(\tau u, (\tau - 1)u)$$

- ▶ u : actual minus forecast quantile; τ : quantile level; coverage of an 80% interval: share of actuals between the 10% and 90% quantiles

- M4 measures [Makridakis, Spiliotis and Assimakopoulos \(2020\)](#)

$$\text{sMAPE} = \frac{200}{H} \sum_{h=1}^H \frac{|y - \hat{y}|}{|y| + |\hat{y}|}$$

- ▶ H : horizon; MASE: MAE divided by the in-sample MAE of the seasonal naive [Hyndman and Koehler \(2006\)](#); OWA: the mean of both ratios to Naive 2

Prerequisites for Today (8/8): tests and the load benchmark

- DM test with the HLN correction and $h - 1$ lags (Chapter 1) Diebold and Mariano (1995): t statistic of the mean loss difference; MCS Hansen, Lunde and Nason (2011): the set of models not significantly worse than the best
- Expert ARX for load (Chapter 1) Ziel and Weron (2018): one regression per hour on the loads of days d , $d - 1$, $d - 6$, the minimum, maximum and last hour of day d , weekday dummies

A1: lasso and ridge under an orthonormal design [Solved]

Task

- ▣ OLS coefficients $z = (2.0, -0.6, 0.3, 1.2)$, $X'X = nI$, $\lambda = 0.5$.
- ▣ 1. Show that the lasso problem separates into $\frac{1}{2}(\beta_j - z_j)^2 + \lambda|\beta_j|$ and derive the soft-thresholding solution.
- ▣ 2. Compute the lasso and the ridge coefficients.
- ▣ 3. Which coefficients does each method set to zero?
- ▣ Report: one derivation; eight numbers; one sentence.

Solution

- ▣ 1. $\|y - X\beta\|^2/(2n) = \text{const} + \frac{1}{2}\|\beta - z\|^2$; subgradient $0 \in \beta_j - z_j + \lambda\partial|\beta_j|$ gives $\beta_j = 0$ if $|z_j| \leq \lambda$, else $z_j - \lambda \text{sign}(z_j)$
- ▣ 2. Lasso: 1.50, -0.10, 0.00, 0.70; ridge: 1.333, -0.400, 0.200, 0.800
- ▣ 3. Lasso zeroes the third coefficient; ridge zeroes none
- ▣ Interpretation: lasso subtracts the same λ from every coefficient, so large coefficients are biased by λ ; ridge shrinks proportionally and never selects

A2: adaptive lasso and elastic net [Proposed]

Solution discussed in class

Task

- ▣ The same z and $\lambda = 0.5$. Model: A1.
- ▣ 1. Adaptive lasso with weights $w_j = 1/|z_j|$ ($\gamma = 1$): compute the four thresholds and the coefficients.
- ▣ 2. Elastic net with $\alpha = 0.5$: compute $S(z_j, \lambda\alpha)/(1 + \lambda(1 - \alpha))$.
- ▣ 3. Compare the bias on the largest coefficient across lasso, adaptive lasso and elastic net.
- ▣ Report: eight numbers; four numbers; one sentence.

A3: the variance of bagging [Solved]

Task

- ▣ B trees, each with variance $\sigma^2 = 1$, pairwise correlation $\rho = 0.3$.
- ▣ 1. Derive $\text{Var}(\frac{1}{B} \sum_b \hat{f}_b) = \rho\sigma^2 + (1 - \rho)\sigma^2/B$.
- ▣ 2. Compute it for $B = 1, 10, 100$ and its limit.
- ▣ 3. Which B brings the variance within 5% of its limit?
- ▣ Report: one derivation; four numbers; one number.

Solution

- ▣ 1.
$$\frac{1}{B^2} [\sum_b \text{Var} \hat{f}_b + \sum_{b \neq c} \text{Cov}(\hat{f}_b, \hat{f}_c)] = \frac{1}{B^2} [B\sigma^2 + B(B-1)\rho\sigma^2]$$
- ▣ 2. 1.000, 0.370, 0.307; limit 0.30
- ▣ 3. $(1 - \rho)/B \leq 0.05\rho \Rightarrow B \geq 47$
- ▣ Interpretation: beyond a few dozen trees only decorrelation (feature subsampling in random forests) lowers the variance

A4: boosting as gradient descent [Proposed]

Solution discussed in class

Task

- ▣ Squared loss $\ell = \frac{1}{2}(y - F)^2$, shrinkage $\nu = 0.1$, base learner: a constant (a tree without splits), $F_0 = 0$. Model: A3.
- ▣ 1. Show that the negative gradient is the residual and that $F_M = (1 - (1 - \nu)^M)\bar{y}$.
- ▣ 2. Compute $F_M/\bar{y}$ for $M = 10$ and 50 , and the M needed to reach 90%.
- ▣ 3. Write the negative gradient of the pinball loss $\rho_\tau(y - F)$.
- ▣ Report: one derivation; three numbers; one formula.

A5: gradients through a scalar RNN [Solved]

Task

- ▣ $h_t = \tanh(wh_{t-1} + ux_t)$ with $w = 0.9$.
- ▣ 1. Show that $\partial h_T / \partial h_{T-k} = \prod_{j=0}^{k-1} w(1 - h_{T-j}^2)$ and that $|\partial h_T / \partial h_{T-k}| \leq |w|^k$.
- ▣ 2. Compute the bound for $k = 10$ and $k = 50$.
- ▣ 3. If $|h_t| = 0.5$ along the path, compute the actual factor per step and the gradient at $k = 10$ and 50.
- ▣ Report: one derivation; two numbers; three numbers.

Solution

- ▣ 1. Chain rule: $\partial h_t / \partial h_{t-1} = w \tanh'(\cdot) = w(1 - h_t^2)$, and $0 < 1 - h_t^2 \leq 1$
- ▣ 2. $0.9^{10} = 0.349$, $0.9^{50} = 0.00515$
- ▣ 3. Factor 0.675; 0.0196 at $k = 10$, 2.9×10^{-9} at $k = 50$
- ▣ Interpretation: saturation of the units speeds up the decay; a signal 50 steps back cannot move the weights

A6: the LSTM forget gate [Proposed]

Solution discussed in class

Task

- ▣ LSTM cell path $c_t = f_t c_{t-1} + i_t \tilde{c}_t$, with a constant forget gate $f = \sigma(b_f)$. Model: A5.
- ▣ 1. Show that the direct path gives $\partial c_T / \partial c_{T-k} = f^k$.
- ▣ 2. Compute f for $b_f = 0, 1, 3$ and f^{50} in each case.
- ▣ 3. Why does the default initialisation not protect against vanishing gradients?
- ▣ Report: one derivation; six numbers; one sentence.

A7: attention by hand [Solved]

Task

- ▣ Keys $k_1 = (1, 0)$, $k_2 = (0, 1)$, $k_3 = (1, 1)$; values $v = (1, 2, 3)$; query $q = (1, 1)$; $d_k = 2$.
- ▣ 1. Compute the scores $qk_j / \sqrt{d_k}$.
- ▣ 2. Compute the softmax weights and the output.
- ▣ 3. What happens to the weights if the three tokens are permuted?
- ▣ Report: three numbers; four numbers; one sentence.

Solution

- ▣ 1. 0.707, 0.707, 1.414
- ▣ 2. Weights 0.248, 0.248, 0.503; output 2.255
- ▣ 3. The weights are permuted with the tokens and the output is unchanged: attention ignores order unless positions are encoded
- ▣ Interpretation: the critique of Zeng et al. starts here: a time series is an ordered object

A8: DLinear is one linear map [Proposed]

Solution discussed in class

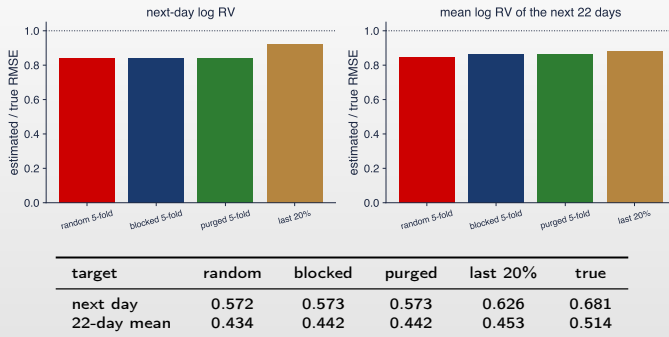
Task

- ▣ $L = 3$ inputs, one output, moving average of length 3 with replicate padding; $W_s = (0.2, 0.3, 0.5)$, $W_t = (0.1, 0.1, 0.8)$. Model: A7.
- ▣ 1. Write the 3×3 matrix A with $\text{trend} = Ax$.
- ▣ 2. Show that the forecast is Wx with $W = W_s(I - A) + W_tA$ and compute W .
- ▣ 3. What does the result imply for the expressiveness of DLinear?
- ▣ Report: a matrix; three numbers; one sentence.

B1: validating HAR on FTSE 100 realised variance [Solved]

- **Question:** do random and blocked cross-validation estimate the future error of HAR, for one-day and 22-day targets?
- **Inputs:** Oxford-Man 5-minute RV of the FTSE 100, $n = 5\,543$ rows; in-set: the first 80% (to 5 September 2017); HAR on $\log RV$
- **Tasks**
 1. Estimate the RMSE on the in-set by random 5-fold, blocked 5-fold, blocked 5-fold with a purge of h rows, and the last 20% of the in-set.
 2. Compute the true RMSE: the model fitted on the in-set, evaluated on the last 20% of the sample.
 3. Repeat for the target “mean $\log RV$ of the next 22 days”.
 4. Interpretation: which scheme would you trust, and why are all of them below the true error?
- **Report:** ten RMSE values, one sentence
- **Notebook:** section B1

B1: solution [Solved]



- Next day: random and blocked agree (HAR errors are nearly uncorrelated, the Bergmeir–Hyndman–Koo case); 22-day mean: random is lower (0.434 against 0.442): overlapping targets leak
- Interpretation: every scheme misses the 2020 volatility episode of the evaluation period: CV estimates the risk of the past regime; the last block is the most honest

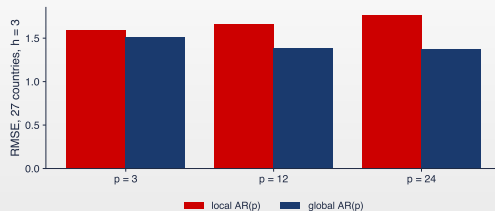
B2: a random forest with a time index [Proposed]

- **Question:** how much does a time-index feature inflate the optimism of random cross-validation?
- **Inputs:** DAX log RV (Oxford-Man); random forest (100 trees) on the 22 lags; target: mean log *RV* of the next 22 days; model: B1
- **Tasks**
 1. Compute the four CV estimates and the true RMSE with the lags only.
 2. Add the row number as a feature and repeat.
 3. Interpretation: which scheme is fooled by the time index, and which is not?
- **Report:** ten RMSE values, one sentence
- **Notebook:** section B2

B3: local and global AR for EU inflation [Solved]

- **Question:** does one AR model for all 27 countries forecast annual inflation three months ahead better than 27 separate models?
- **Inputs:** Eurostat HICP annual rates, 27 EU countries, rolling 120-month windows, 137 origins from January 2015; direct forecasts $h = 3$
- **Tasks**
 1. Fit local $AR(p)$ (one OLS per country) and global $AR(p)$ (one pooled OLS) for $p = 3, 12, 24$.
 2. Compute the RMSE over the 27 countries and for Romania.
 3. Compute the DM statistic of global against local on the cross-country average loss.
 4. Interpretation: why does the gap widen with p ?
- **Report:** twelve RMSE values, three DM statistics, one sentence
- **Notebook:** section B3

B3: solution [Solved]



p	local	global	DM (27)	p -value	local RO	global RO
3	1.589	1.514	-1.56	0.121	1.463	1.498
12	1.660	1.386	-2.29	0.024	1.327	1.373
24	1.766	1.376	-3.11	0.002	1.395	1.372

- Interpretation: local AR deteriorates with p (about 100 rows for $p + 1$ coefficients), global AR improves (2700 rows); at $p = 24$ the global gain is significant ($p = 0.002$); for Romania alone the difference is not significant

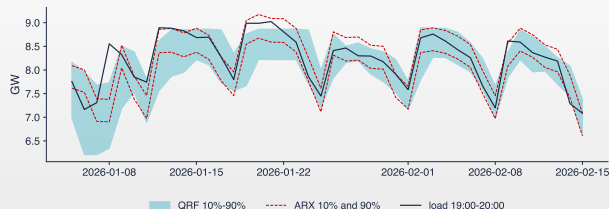
B4: the panel lasso for Romania [Proposed]

- **Question:** do the inflation rates of the other 26 EU countries help forecast Romanian inflation three months ahead?
- **Inputs:** 3 own lags (not penalised) and 3 lags of the 26 other countries; rolling 120-month windows, refit every 3 months; penalty by blocked 5-fold CV with a purge of 3 months; model: B3
- **Tasks**
 1. Forecast with AR(3), ridge, lasso, adaptive lasso, elastic net, post-lasso OLS and the global AR(12).
 2. Compute the RMSE and the DM test of each model against AR(3).
 3. Report the median number of selected predictors and the three most selected countries.
 4. Interpretation: is the high-dimensional panel worth it here?
- **Report:** seven RMSE values, six DM p -values, the selection summary, one sentence
- **Notebook:** section B4

B5: QRF and ARX for the evening peak [Solved]

- **Question:** does a quantile regression forest give better day-ahead quantiles than the expert ARX for the load at 19:00–20:00?
- **Inputs:** Romanian hourly load, hour 19:00–20:00; 638 forecast days from January 2025; QRF (200 trees, leaves of at least 5) refitted monthly on all earlier data; ARX with empirical error quantiles ([Chapter 1](#))
- **Tasks**
 1. Compute the deciles of both methods for each day.
 2. Compute the average pinball loss, the 80% coverage, the mean width of the 80% interval and the MAE of the point forecast.
 3. Test the difference of the daily pinball losses with DM.
 4. Interpretation: which bands would a grid operator prefer?
- **Report:** eight numbers, one DM statistic, one sentence
- **Notebook:** section B5

B5: solution [Solved]



	pinball (MW)	coverage 80%	width (MW)	MAE (MW)
QRF	78.5	86.7%	784	195
ARX	77.6	77.6%	530	187

- DM on the pinball difference: 0.40 ($p = 0.69$): no significant difference
- Interpretation: QRF bands are wider and over-cover (86.7%), ARX bands are sharper and slightly under-cover; equal scores, so the choice depends on the operator's cost of a shortfall

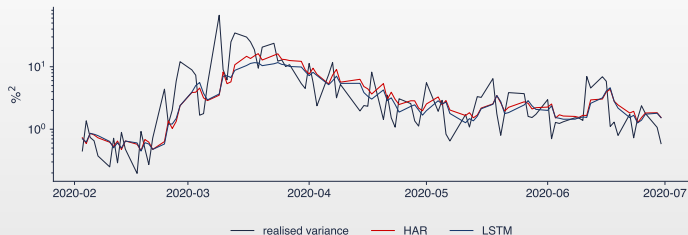
B6: monotone boosting for the evening peak [Proposed]

- **Question:** does forcing the forecast to increase with the lagged loads cost accuracy at the evening peak?
- **Inputs:** the same rows as B5; histogram gradient boosting (300 iterations, 15 leaves), with and without increasing constraints on the three load lags, refitted monthly; model: B5
- **Tasks**
 1. Compute the MAE of both models and the DM test of their absolute errors.
 2. Count the decreasing steps of the partial dependence on the load of day d in the first fit.
 3. Interpretation: is the constraint worth its cost here?
- **Report:** two MAE values, one DM statistic, two counts, one sentence
- **Notebook:** section B6

B7: LSTM against HAR for the FTSE 100 [Solved]

- **Question:** does an LSTM on the last 22 days forecast next-day realised variance better than HAR?
- **Inputs:** FTSE 100 log RV; expanding window from 2500 days, re-estimation every 500 days; forecasts 8 December 2009 – 25 February 2022 ($T = 3\,085$); LSTM with 16 units, average of three seeds
- **Tasks**
 1. Forecast RV_{t+1} with HAR and the LSTM, with the lognormal correction.
 2. Compute the average QLIKE of both and their ratio.
 3. Compute the DM statistic.
 4. Interpretation: would you replace HAR?
- **Report:** two losses, one ratio, one DM statistic, one sentence
- **Notebook:** section B7

B7: solution [Solved]



- ▣ QLIKE: HAR 0.2486, LSTM 0.2474; ratio 0.995; DM -0.63 ($p = 0.53$)
- ▣ Interpretation: the LSTM is indistinguishable from HAR; it costs seeds, tuning and minutes of training for no measurable gain: keep HAR, or average the two

B8: DLinear and N-BEATS on 100 M4 hourly series [Proposed]

- **Question:** does a global N-BEATS still beat the benchmarks when it is trained on only 100 series?
- **Inputs:** the first 100 M4 hourly series, horizon 48, input 336 hours, one seed; model: B7
- **Tasks**
 1. Forecast with Naive 2, the seasonal naive, global DLinear and global N-BEATS.
 2. Compute sMAPE, MASE and OWA.
 3. Interpretation: compare with the lecture, where N-BEATS used all 414 series and three seeds.
- **Report:** twelve numbers, one sentence
- **Notebook:** section B8

C1: deep learners against HAR for crypto-assets [Proposed]

- **Question:** is crypto volatility, with its 24/7 trading and frequent regime changes, a place where learners beat HAR?
- **Inputs:** Bitcoin and Ether daily 5-minute RV from Binance one-minute prices, 2018–2026; expanding window from 1000 days, refit every 250 days; model: B7
- **Tasks**
 1. Write a five-line pre-registration: hypothesis, learners, horizons, loss and test, what would change your mind.
 2. Forecast with HAR, an MLP and boosting at $h = 1$ and $h = 5$; compute QLIKE ratios and DM tests against HAR.
 3. Interpretation: what do the data support, and how would you extend the study?
- **Report:** a five-line plan, eight ratios with p -values, a project plan
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise machine learning for time series. The answer:
- ▣ (a) Random K-fold cross-validation is never valid for time series.
- ▣ (b) A TCN with kernel 3 and dilations 1, 2, 4, 8 (two convolutions per block) sees the last 31 hours.
- ▣ (c) With enough trees, bagging drives the variance of the forecast to zero.
- ▣ (d) LSTMs never suffer from vanishing gradients, thanks to their gates.
- ▣ (e) Zeng et al. (2023) proved that every Transformer is worse than a linear model.
- ▣ (f) The patch with the largest attention weight is the one the forecast depends on most.
- ▣ (g) Shapley values of all features sum to the prediction minus the average prediction over the background.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ Validate with blocks and purges when targets overlap or features carry time; random K-fold only for well-specified autoregressions
- ▣ Pooling series (global models) buys memory; selection on short panels is unstable
- ▣ Ensembles: the variance floor is $\rho\sigma^2$; boosting fits gradients; constraints must be right everywhere
- ▣ Gates and attention have simple algebra: forget gates must be open, attention ignores order
- ▣ Strong baselines (HAR, expert ARX, seasonal naive) are hard to beat; test every claim

After the seminar

- Lecture 12 develops each topic of today: learning theory for dependent data, high-dimensional inference, tree ensembles, recurrent and attention models, N-BEATS, DeepAR and honest benchmarking
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: learners against HAR for crypto-assets, pre-registered
- Reading: [Bergmeir, Hyndman and Koo \(2018\)](#); [Montero-Manso and Hyndman \(2021\)](#); [Zeng et al. \(2023\)](#); [Oreshkin et al. \(2020\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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